

2107 Long Question

1. Explain the arrangement of tissues in dicotyledonous stem.
 - The transverse section of a typical young dicotyledonous stem shows that the epidermis is the outermost protective layer of the stem.
 - Covered with a thin layer of cuticle, it may bear trichomes and a few stomata.
 - The cells arranged in multiple layers between epidermis and pericycle constitutes the cortex.
 - It consists of three sub-zones. The outer hypodermis consists of a few layers of collenchymatous cells just below the epidermis, which provide mechanical strength to the young stem.
 - Cortical layers below hypodermis consist of rounded thin walled parenchymatous cells with conspicuous intercellular spaces.
 - The innermost layer of the cortex is called the endodermis. The cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath.
 - Pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.
 - In between the vascular bundles there are a few layers of radially arranged parenchymatous cells, which constitute medullary rays.
 - A large number of vascular bundles are arranged in a ring; the 'ring' arrangement of vascular bundles is a characteristic of dicot stem.
 - Each vascular bundle is conjoint, open, and with endarch protoxylem. A large number of rounded, parenchymatous cells with large intercellular spaces which occupy the central portion of the stem constitute the pith.

Bot 2107 Short Questions

1. Clarify the organismal theory.
 - By the latter half of the nineteenth century, an alternative to the cell theory was formulated.
 - It was known as the organismal theory and it replaced some of the idea expounded by the cell theory.
 - The proponents of the organismal theory consider the entire organism, rather than the individual cells, to be the prime importance.
 - The many-celled plant or animal is regarded not merely as a group of independent units, but as a more or less continuous mass of protoplasm, which, in the course of evolution, become subdivided into cells.
 - The organismal theory is especially applicable to plants, whose protoplasts do not pinch apart during cell division.

2. Give the aleurone grains with relevant diagram.

- Each aleurone grain is a solid, ovate or rounded body, encloses a crystal-like body in it, known as crystalloid and a globule like body known as globoid.
- The crystalloid is proteinaceous in nature and occupies the major portion lot of aleurone grains.
- The globoid is double phosphate of calcium (Ca) and magnesium (Mg) and occupies the narrower part of the grain.
- The aleurone grains vary in their shape and size in different plants.

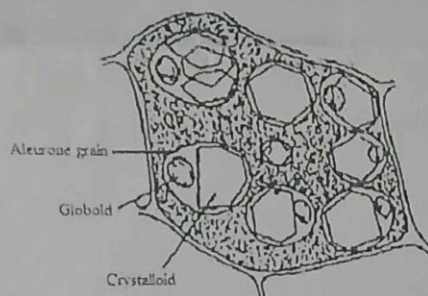


Fig. 2.10 Diagram showing Aleurone grains

3. Write short notes on stomata as found in epidermis.

- Stomata are structures present in the epidermis of leaves. Stomata regulate the process of transpiration and gaseous exchange.
- Each stoma is composed of two bean shaped cells known as guard cells. In grasses, the guard cells are dumb-bell shaped.
- The outer walls of guard cells (away from the stomatal pore) are thin and the inner walls (towards the stomatal pore) are highly thickened.
- The guard cells possess chloroplasts and regulate the opening and closing of stomata.
- Sometimes, a few epidermal cells, in the vicinity of the guard cells become specialized in their shape and size and are known as subsidiary cells.
- The stomatal aperture, guard cells and the surrounding subsidiary cells are together called stomatal apparatus.